

LC 3144: Minimum Substring Partition of Equal Character Frequency

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Problem Understanding

Tags: Amazon, Microsoft

need to partition s to make 1 or more balanced substring

"**Balanced string**" = a string where each character in string occurs same number of times

Precomputation Logic

a → 2
b → 2
c → 3

→ now we will precompute if $i...j$ is balanced or not

by doing this we won't have to compute everytime for each block $[i...j]$

Validation Check

say $i...j$ has a → 2, b → 2, c → 3

- total = $(j - i + 1) = 7$
- and $(map.size) = 3$, one frequency = 2

- $\therefore 3 \times 2 = 6$, and $(6 \neq 7)$

it means it won't be balanced

Partition DP

after that, it's simple partition DP.

```
[ 0 --- j-1 ] [ j --- i ]
```

If 0 to j-1 is not valid we continue

C++ Solution

```
class Solution {
public:
    int minimumSubstringsInPartition(string s) {

        int n = s.size();
        // isbal[l][r] to precompute if l...r is balanced or not

        vector<vector<bool>> isbal(n, vector<bool>(n, false));
        for(int l=0; l<n; l++){
            unordered_map<char, int> freq;

            for(int r=l; r<n; r++){
                freq[s[r]]++;

                int f1 = freq[s[l]];
                bool flag = true;
                for(auto &it : freq){
                    if(it.second != f1){
                        flag = false;
                        break;
                    }
                }
            }
        }
    }
}
```

```
        if(flag){
            isbal[l][r] = true;
        }
    }
}

vector<int> dp(n+1, INT_MAX);
dp[0] = 0;
for(int i=1; i<=n; i++){
    for(int j=i-1; j>=0; j--){
        // [0.....j-1][j.....i-1]

        if(isbal[j][i-1] == false) continue;
        dp[i] = min(dp[i], dp[j]+1);
    }
}

return dp[n];
}

};
```